EXAM 2 REVIEW

BLAKE FARMAN

Lafayette College

Name: Solutions

Implicit Differentiation and Related Rates

1 (15 Points)**.** Find dy/dx:

$$\cos(xy) = 1 + \tan y$$

$$\frac{d}{dx}\cos(xy) = -\sin(xy)\frac{d}{dx}(xy)$$

$$= -\sin(xy)(y + x\,dy/dx)$$

$$= -y\sin(xy) - x\sin(xy)\,dy/dx$$

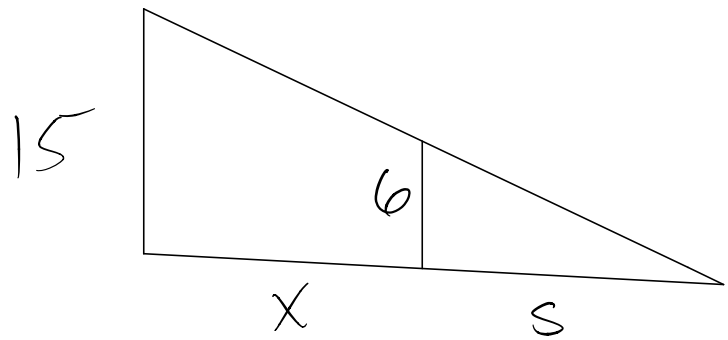
$$= \frac{d}{dx}(1 + \tan(y)) = 0 + \sec^2(y)\,dy/dx$$

$$\Rightarrow -y\sin(xy) = [x\sin(xy) + \sec^2(y)]\,dy/dx$$

$$\Rightarrow \boxed{dy/dx = \frac{-y\sin(xy)}{x\sin(xy) + \sec^2(y)}}$$

Date: March 29, 2019.

2 (20 Points)**.** A street light is mounted at the top of a 15-ft-tall pole. A person 6 ft tall walks away from the pole with a speed of 5 ft/s along a straight path. How fast is the tip of their shadow moving when they are 40 ft from the pole?
(Hint: The length of the shadow is measured from the person to the tip of the shadow; the rate at which the tip of the shadow is moving is measured from the pole to the tip of the shadow.)



$$\frac{x+s}{15} = \frac{s}{6} \Rightarrow 6x+6s = 15s$$

$$\Rightarrow 6x = 9s$$

$$\Rightarrow s = \frac{6x}{9} = \frac{2x}{3}$$

$$\Rightarrow \frac{ds}{dt} = \frac{2}{3}\frac{dx}{dt}$$

$$\Rightarrow \frac{d}{dt}(x+s) = \frac{dx}{dt} + \frac{ds}{dt} = \frac{dx}{dt} + \frac{2}{3}\frac{dx}{dt}$$

$$= \frac{5}{3}\frac{dx}{dt} = \boxed{\frac{25}{3}}$$

Derivatives and Shape

Use the function

$$f(x) = \frac{x^2}{x-1}$$

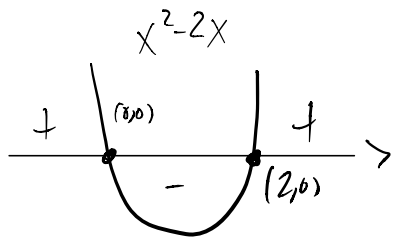
to answer the following questions.

3. Find the intervals where the function is increasing and decreasing. Write NONE if not applicable.

Increasing: $(-\infty, 0) \cup (2, \infty)$

Decreasing: $(0, 2)$

$$f'(x) = \frac{2x(x-1) - x^2(1)}{(x-1)^2} = \frac{2x^2 - 2x - x^2}{(x-1)^2} = \frac{x^2 - 2x}{(x-1)^2}$$



4. State the local maximum and local minimum value(s). Write NONE if not applicable.

Local maximum value(s): $(0, 0)$

Local minimum value(s): $(2, 4)$

$$f(0) = \frac{0^2}{0-1} = 0$$

$$f(2) = \frac{2^2}{2-1} = 4$$

5. Find the intervals on which the function is concave up and concave down. State the inflection points. Write NONE if not applicable.

Concave Up: $(-\infty, 1)$

Concave Down: $(1, \infty)$

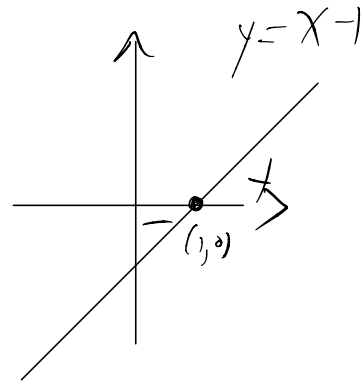
Inflection Points: NONE

$$f''(x) = \frac{(2x-2)(x-1)^2 - (x^2-2x)(2)(x-1)(1)}{(x-1)^4}$$

$$= \frac{(2x-2)(x-1) - 2(x^2-2x)}{(x-1)^3}$$

$$= \frac{2x^2-2x-2x+2-2x^2+4x}{(x-1)^3}$$

$$= \frac{2}{(x-1)^3}$$



Asymptotes

6. Find the asymptotes of

$$f(x) = \frac{x^2}{x-1}$$

Write NONE if there are none.

Horizontal: NONE

Oblique: $y = x+1$

Vertical: $x = 1$

$$x^2 = (x+1)(x-1) + 1$$

$$\Rightarrow \frac{x^2}{x-1} = x+1 + \frac{1}{x-1}$$

Curve Sketching

7 (20 Points)**.** Sketch the curve

$$f(x) = \frac{x^2}{x-1}.$$

(a) State the domain of f.

$x \neq 1$

(b) Find the intercepts and express them as an (x, y) pair. Write NONE if there are none.

x-intercepts: $(0,0)$

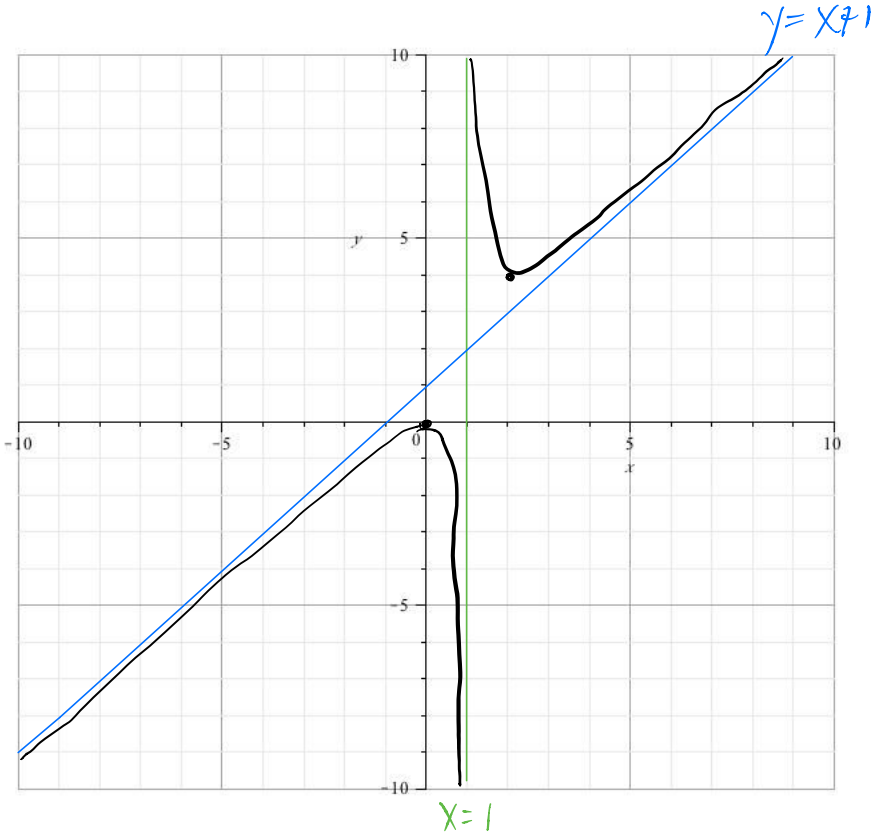
y-intercept: $(0,0)$

(c) Is the function even, odd, or neither? What type of symmetry does the function have?

$$f(-x) = \frac{(-x)^2}{-x-1} = \frac{x^2}{-x-1} \neq f(x), -f(x)$$

So neither.

(d) Sketch the curve. Be sure that your graph is labeled and neat. Messy/incoherent graphs will receive zero points.



Closed Interval Method and Optimization

8 (15 Points)**.** Find the absolute maximum and minimum values of $f(x) = 2x^3 - 3x^2 - 12x + 1$ on $[-2, 3]$.

$f'(x) = 6x^2 - 6x - 12 = 6(x^2 - x - 2) = 6(x-2)(x+1)$

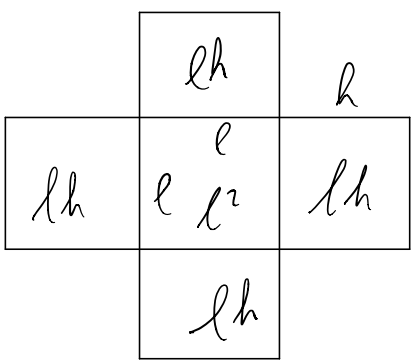
$f(-2) = -16 - 12 + 24 + 1 = -16 + 12 + 1 = 5$

$f(-1) = -2 \cdot 3 + 12 + 1 = \boxed{8}$ max

$f(2) = 16 - 12 - 24 + 1 = 4 - 24 + 1 = -20 + 1 = \boxed{-19}$ min

$f(3) = 54 - 27 - 36 + 1 = 27 - 36 + 1 = -9 + 1 = -8$

9 (10 Points)**.** A box with a square base and open top must have a volume of 32 cm^3. Find the dimensions of the box that minimize the amount of material used.



$V = l^2 h = 32$

$\Rightarrow h = 32/l^2$

$S = l^2 + 4lh = l^2 + 128/l$

$S' = 2l - 128/l^2 = 0$

$\Rightarrow 2l^3 - 128 = 0$

$\Rightarrow l^3 = 128/2 = 64 = 8^2 = 4^3$

$\Rightarrow \boxed{l = 4}$

$\Rightarrow \boxed{h = \frac{32}{16} = 2}$